ModelIt!

Student Activity Pack

Your Inner Fish

Why You Share Bones with a 375-Million-Year-Old Fish

Grade 7 | NGSS MS-LS4-1, MS-LS4-2 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is a fossil and how do you think they form?

2 Why do you think your arm has the same bones as a whale's flipper?

3 Draw what you think a transitional fossil between a fish and a land animal might look like.

4 How do scientists figure out which animals are related to each other?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Environmental Change Rate | Time (Millions of Years) | Anatomical Similarity | Species Diversity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Stable Environment

Low environmental change, long time period

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Mass Extinction

Environmental Change Rate at maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Gradual Change

Moderate change over a long time period

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Mutation Rate — How quickly random genetic changes occur in populations, creating variation

Geographic Isolation — When populations are separated by physical barriers, preventing gene flow

Ecological Niche Availability — The number of open habitats and food sources available for new species

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do humans have the same bones as a fish that lived 375 million years ago? Use evidence from your simulation to support your answers.

1. **How does finding the same bone structure in fish, frogs, birds, and humans support evolution?**

2. **Why are transitional fossils like Tiktaalik so important to scientists?**

3. **What would happen to the fossil record if a massive asteroid hit Earth today?**

4. **How is your model similar to and different from real evolution?**

5. **If evolution is real, why do we still have fish?**

8 STEM Challenge: Build an Evolutionary Evidence Wall

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YOUR ENGINEERING MISSION

Create a visual display showing evidence for evolution from the fossil record, including homologous structures and transitional fossils, for a natural history museum.

THE SCENARIO

A natural history museum needs a new exhibit showing how scientists use fossils and anatomy to trace evolutionary relationships. Your team will design the exhibit using evidence from your model.

Engineering Design Thinking

1. WHICH TRANSITIONAL FOSSILS WILL YOU FEATURE AND WHY?
2. HOW WILL YOU SHOW HOMOLOGOUS STRUCTURES BETWEEN FISH, REPTILES, AND MAMMALS?
3. HOW WILL YOU REPRESENT THE TIMELINE OF EVOLUTION VISUALLY?
4. WHAT INTERACTIVE ELEMENTS WILL HELP VISITORS UNDERSTAND THE EVIDENCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Paleontologists excavate and study fossils to understand the history of life on Earth. They work for museums, universities, and geological surveys, earning $55,000–$110,000/year.